Andreas Tzitzikas

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Education

University of Maryland, College Park, MD — Expected May 2027

Master of Science in Computer Engineering — **GPA: 4.00**

Bachelor of Science in Computer Engineering (GPA: 3.88) — Expected May 2026

Relevant Coursework: Digital Computer Design (ENEE646), Digital CMOS VLSI Design (ENEE640), Modern Digital System Design (ENEE408C), Parallel Computing (CMSC416), Computer Organization (ENEE350), Algorithms (CMSC351)

Projects

Tomasulo Out-of-Order RISC-V CPU | SystemVerilog, OpenLane ASIC Flow — Nov 2025 - Dec 2025

- Architected complete out-of-order CPU using **Tomasulo's Algorithm** with register renaming, 8-entry reservation stations, 16-entry reorder buffer, and common data bus for dynamic scheduling and speculative execution.
- Achieved **36% higher frequency** (69.7MHz vs 51.2MHz) and 40-150% IPC improvement on complex workloads compared to in-order implementation through comprehensive benchmarking and performance analysis.
- Synthesized to fabrication-ready GDSII using **Sky130 130nm PDK**, achieving 0 DRC violations, 0 LVS errors, and meeting all PPA constraints with comprehensive static timing analysis.
- Implemented sophisticated **branch prediction and recovery** mechanisms, dynamic hazard resolution, and in-order commit ensuring precise exceptions and program correctness.

5-Stage In-Order RISC-V CPU | SystemVerilog, ASIC Implementation — Aug 2025 - Nov 2025

- Designed and implemented complete 5-stage pipelined CPU supporting full **RV32I ISA** with hazard detection, data forwarding, and comprehensive branch/jump support across 10 RTL modules.
- Achieved **100% functional verification** through extensive testbench development and automated testing framework, validating all instruction types and pipeline corner cases.
- Synthesized to **10,121 standard cells** with 1.0mm^2 die area using OpenLane flow, achieving 50MHz operation with zero timing violations and DRC/LVS clean layout.
- Optimized critical path timing through strategic pipeline balancing, synchronous reset implementation, and comprehensive static timing analysis for ASIC manufacturability.

Polynomial Evaluation Accelerator | SystemVerilog, FPGA Synthesis — Fall 2025

- Architected high-throughput accelerator using LIDE-V dataflow paradigm, implementing 5 distinct finite state machines for FIFO handshaking and resource arbitration across 9 processing modules.
- Achieved 100% functional coverage through comprehensive SystemVerilog testbench development with 15+ complex test cases validated against C reference models.
- Optimized memory access with dual-port coefficient memory and auto-padding logic, enabling batch processing of up to 31 polynomial evaluations with minimal latency.

Experience

Embedded Systems Intern, Alchemity – College Park, MD — Jun 2025 – Aug 2025

- Accelerated validation of a hybrid solid oxide fuel cell reactor by developing a full-stack automation suite (**Rust**, **Python**) to parallelize testing, converting multi-day manual experiments into overnight runs to significantly increase throughput.
- Engineered real-time **STM32** firmware to precisely control material synthesis within modular reactors, featuring non-blocking motor/relay drivers and **SPI** peripheral management.
- Built end-to-end control interfaces including desktop GUIs, embedded display drivers, and CLI tools to streamline workflows between hardware and software teams.

Technical Coordinator, Electronics Prototyping Lab, Terrapin Works – College Park, MD — Jan 2024 – Present

- Provide end-to-end **PCB** support (schematic to assembly) for 20+ students and faculty per semester, maintaining **LPKF** tools to ensure **90% uptime**.
- Lead training for new employees and expand campus outreach to increase awareness of PCB services, while assisting researchers with diagnostics.

Skills

HDLs & Languages: SystemVerilog, Verilog, C/C++, Python, Tcl, Bash
Computer Architecture: RISC-V ISA, Tomasulo's Algorithm, Pipelined Datapaths, Out-of-Order Execution, Hazard Control, RTL Design
ASIC/FPGA Tools: OpenLane, Yosys, Sky130 PDK, Xilinx Vivado, Icarus Verilog, Verible, GTKWave, Static Timing Analysis, DRC/LVS Verification
Methodologies: PPA Optimization, Testbench Development, Functional Verification, Git, Linux